

10-Mark Questions & Answers — MPCA Module 5: 80386DX & Pentium Processors

Q1. Explain 80386DX architecture, register organization, operating modes, memory management (Segmentation & Paging), and Pentium superscalar pipelining in detail.

Comprehensive master answer covering:

1. 80386DX 32-bit block diagram and functional units.
2. Extended registers (EAX-ESP, EFLAGS, CR0-CR3).
3. Real Mode vs Protected Mode (Privilege rings 0-3, GDT/LDT descriptors) vs Virtual 8086 Mode.
4. Two-level 4KB Paging mechanism.
5. Pentium Superscalar U & V pipelines, Branch Target Buffer (BTB), and MESI cache protocol.